

Neuronal Activity, Mitogens, and mTOR: Overcoming the Hurdles for the Treatment of Glioblastoma Multiforme.

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Glioblastoma multiforme (GBM) and other malignant gliomas are considered to be the most prevalent of primary malignant brain tumors. The incidence of these tumors per year is reported as 4.13 per 100,000 individuals per year. The median survival time following the diagnosis of GBM is approximately fifteen months in the setting of providing presently available treatments with surgical resection, radiation, and chemotherapy. Given these statistics, new strategies for the treatment of GBM and other aggressive tumors of the brain are warranted.

The Septin Cytoskeleton is Required for Plasma Membrane Repair.

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Mammalian cells are frequently exposed to mechanical and biochemical stressors resulting in plasma membrane injuries. Repair mechanisms reseal the plasma membrane to restore homeostasis and prevent cell death. In the present work, a silencing RNA screen was performed to uncover plasma membrane repair mechanisms of cells exposed to a pore-forming toxin (listeriolysin O). This screen identified molecules previously known to repair the injured plasma membrane such as annexin A2 (ANXA2) as well as novel plasma membrane repair candidate proteins. Of the novel candidates, we focused on septin 7 (SEPT7) because the septins are an important family of conserved eukaryotic cytoskeletal proteins. Using diverse experimental approaches, we established for the first time that SEPT7 plays a general role in plasma membrane repair of cells perforated by pore-forming toxins and mechanical wounding. Remarkably, upon cell injury, the septin cytoskeleton is extensively redistributed in a Ca^{2+} -dependent fashion, a hallmark of plasma membrane repair machineries. The septins reorganize into subplasmalemmal domains arranged as knob and loop (or ring) structures containing F-actin, myosin II, and annexin A2 (ANXA2) and protrude from the cell surface. Importantly, the formation of these domains correlates with the plasma membrane repair efficiency. Super-resolution microscopy shows that septins and actin are arranged in intertwined filaments associated with ANXA2. Silencing SEPT7 expression prevented the formation of the F-actin/myosin II/ANXA2 domains, however, silencing expression of ANXA2 had no observable effect on their formation. These results highlight the key structural role of the septins in remodeling the plasma membrane and in the recruitment of the repair molecule ANXA2. Collectively, our data support a novel model in which the septin cytoskeleton acts as a scaffold to promote the formation of plasma membrane repair domains containing contractile F-actin and annexin A2.

Harnessing Microbial Effectors for Macrophage-Mediated Drug Delivery.

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Macrophage-based drug delivery systems are promising, but their development is still in its infancy, with many limitations remaining to be addressed. Our aim was to design a system harnessing microbial effectors to facilitate controlled drug cargo expulsion from macrophages to enable the use of more toxic drugs without adding to the risk of off-target detrimental effects. The pore forming and actin polymerizing *Listeria monocytogenes* effectors listeriolysin-O (LLO) and actin assembly-inducing protein

(ActA) were synthesized using a novel green fluorescent protein (GFP)-linked heterologous expression system. These effectors were coated onto polystyrene beads to generate "synthetic cargo" before loading into primary M1 macrophages. Bead uptake and release from macrophages were evaluated by using high-throughput quantitative imaging flow cytometry and confocal microscopy. In vitro results confirmed appropriate activity of synthesized effectors. Coating of these effector proteins onto polystyrene beads (simulated drug cargo) resulted in changes in cellular morphology, bead content, and intracellular bead localization, which may support an interpretation of the induced release of these beads from the cells. This forms the basis for further investigation to fully elucidate any potential release mechanisms. Bacterial effectors ActA and LLO successfully effectuated actin polarization and protrusions from cell membranes similar to those seen in cells infected with *Listeria* spp., illustrating the potential of using these effectors and production methods for the development of an endogenous drug delivery system capable of low-risk, targeted release of high potency drugs.

PoreGlow: A split green fluorescent protein-based system for rapid detection of *Listeria monocytogenes*.

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Listeria monocytogenes, a severe foodborne pathogen causing severe diseases underscores the necessity for the development of a detection system with high specificity, sensitivity and utility. Herein, the PoreGlow system, based on split green fluorescent protein (GFP), was developed and assessed for the fast and accurate detection of *L. monocytogenes*. Split GFP-encapsulated liposomes were optimized for targeted analysis. The system utilizes listeriolysin O (LLO), a toxin produced by *L. monocytogenes* that enlarges the pores split GFP-encapsulated liposomes, to detect *L. monocytogenes* by measuring the fluorescent signal generated when the encapsulated GFP is released and reacted with the externally added fragment of the split GFP. The system exhibited a limit of detection of 0.17 µg/ml for LLO toxin and 10 CFU/mL for *L. monocytogenes* with high sensitivity and specificity and no cross-reactivity with other bacteria. The PoreGlow system is practical, rapid, and does not require sample pre-treatment, making it a promising tool for the early detection of *L. monocytogenes* in food products, which is crucial for preventing outbreaks and protecting public health.

Nanoelectrospray based synthesis of large, transportable membranes with integrated membrane proteins.

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Membrane proteins tend to be difficult to study since they need to be integrated into a lipid bilayer membrane to function properly. This study presents a method to synthesize a macroscopically large and freely transportable membrane with integrated membrane proteins which is useful for studying membrane proteins and protein complexes in isolation. The method could serve as a blueprint for the production of larger quantities of functionalised membranes for integration into technical devices similar to the MinION DNA sequencer. It is possible to self-assemble larger biological membranes on solid surfaces. However, they cannot be removed from their solid support without destroying them. In transportable form, self-assembled membranes are limited to sizes of about 17 nm in nanodiscs. Here we electrospray a series of molecular layers onto the liquid surface of a buffer solution which creates a flat,

liquid environment on the surface that directs the self-assembly of the membrane. This method enables us to experimentally control the membrane composition and to succeed in producing large membranes with integrated OmpG, a transmembrane pore protein. The technique is compatible with the assembly of membrane based protein complexes. Listeriolysin O and pneumolysin efficiently assemble into non-covalent membrane pore complexes of approximately 30 units or more within the surface layer.

Activation of inflammasomes and mechanisms for intracellular recognition of *Listeria monocytogenes*.

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The high mortality rate associated with *Listeria monocytogenes* can be attributed to its ability to invade the body systemically and to activate inflammasomes. Both of these processes are facilitated by expressing a major virulence factor known as listeriolysin O, a 56 kDa pore-forming protein encoded by the *hly* gene. Listeriolysin O plays a crucial role in the pathogenesis of the bacterium by facilitating the escape of the pathogen from the phagosome into the cytosol. This process is essential for the successful establishment of infection. In addition, listeriolysin O is known as an immunomodulator that activates host signal transduction. In addition to listeriolysin O, *Listeria* expresses a variety of bacterial ligands, such as lipoteichoic acid, nucleotide, and flagellin, that are recognized by host intracellular pattern-recognition receptors including Nod-like receptors, AIM2-like receptors, and RIG-I-like receptors. This review introduces intracellular recognition of *Listeria monocytogenes* since recent studies have revealed that the activation of inflammasome exacerbates Gram-positive bacteria infection.

Relative Roles of Listeriolysin O, InlA, and InlB in *Listeria monocytogenes* Uptake by Host Cells.

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Listeria monocytogenes is a facultative intracellular pathogen that infects a wide variety of cells, causing the life-threatening disease listeriosis. *L. monocytogenes* virulence factors include two surface invasins, InlA and InlB, known to promote bacterial uptake by host cells, and the secreted pore-forming toxin listeriolysin O (LLO), which disrupts the phagosome to allow bacterial proliferation in the cytosol. In addition, plasma membrane perforation by LLO has been shown to facilitate *L. monocytogenes* internalization into epithelial cells. In this work, we tested the host cell range and importance of LLO-mediated *L. monocytogenes* internalization relative to the canonical invasins, InlA and InlB. We measured the efficiencies of *L. monocytogenes* association with and internalization into several human cell types (hepatocytes, cytotrophoblasts, and endothelial cells) using wild-type bacteria and isogenic single, double, and triple deletion mutants for the genes encoding InlA, InlB and LLO. No role for InlB was detected in any tested cells unless the InlB expression level was substantially enhanced, which was achieved by introducing a mutation (*prfA**) in the gene encoding the transcription factor PrfA. In contrast, InlA and LLO were the most critical invasion factors, although they act in a different manner and in a cell-type-dependent fashion. As expected, InlA facilitates both bacterial attachment and internalization in cells that express its receptor, E-cadherin. LLO promotes *L. monocytogenes* internalization into hepatocytes, but not into cytotrophoblasts and endothelial cells. Finally, LLO and InlA cooperate to increase the efficiency of host cell invasion by *L. monocytogenes*.

Role of listeriolysin O in cell-to-cell spread of *Listeria monocytogenes*.

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Listeria monocytogenes is a facultative intracellular bacterial pathogen that escapes from a host vacuolar compartment and grows rapidly in the cytosol. Listeriolysin O (LLO) is a secreted pore-forming protein essential for the escape of *L. monocytogenes* from the vacuole formed upon initial internalization. However, its role in intracellular growth and cell-to-cell spread events has not been testable by a genetic approach. In this study, purified six-His-tagged LLO (HisLLO) was noncovalently coupled to the surface of nickel-treated LLO-negative mutants. Bound LLO mediated vacuolar escape in approximately 2% of the mutants. After 5.5 h of growth, cytosolic bacteria were indistinguishable from wild-type bacteria with regard to formation of pseudopod-like extensions, here termed listeriopods, and spread to adjacent cells. However, bacteria in adjacent cells failed to multiply and were found in double-membrane vacuoles. Addition of bound LLO to mutants lacking LLO and two distinct phospholipases C (PLCs) also resulted in spread to adjacent cells, but these triple mutants became trapped in multiple-membrane vacuoles that are reminiscent of autophagocytic vacuoles. These studies show that neither LLO nor the PLCs are necessary for listeriopod formation and uptake of bacteria into neighboring cells but that LLO is required for the escape of *L. monocytogenes* from the double-membrane vacuole that forms upon cell-to-cell spread.

Listeriolysin O-dependent host surfaceome remodeling modulates *Listeria monocytogenes* invasion.

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Listeria monocytogenes is a pathogenic bacterium that invades epithelial cells by activating host signaling cascades, which promote bacterial engulfment within a phagosome. The pore-forming toxin listeriolysin O (LLO), which is required for bacteria phagosomal escape, has also been associated with the activation of several signaling pathways when secreted by extracellular bacteria, including Ca^{2+} influx and promotion of *L. monocytogenes* entry. Quantitative host surfaceome analysis revealed significant quantitative remodeling of a defined set of cell surface glycoproteins upon LLO treatment, including a subset previously identified to play a role in the *L. monocytogenes* infection process. Our data further shows that the lysosomal-associated membrane proteins LAMP-1 and LAMP-2 are translocated to the cellular surface and those LLO-induced Ca^{2+} fluxes are required to trigger the surface relocalization of LAMP-1. Finally, we identify late endosomes/lysosomes as the major donor compartments of LAMP-1 upon LLO treatment and by perturbing their function, we suggest that these organelles participate in *L. monocytogenes* invasion.

Characterization of *Listeria monocytogenes* pathogenesis in a strain expressing perfringolysin O in place of listeriolysin O.

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Listeriolysin O (LLO) is a pore-forming cytolysin that enables *Listeria monocytogenes* to escape from a

host cell vacuole. The structural gene for the related cytolysin perfringolysin O (pfo) was cloned downstream from the promoter for hly, the gene encoding LLO, both on a plasmid and on the *L. monocytogenes* chromosome. Both strains secreted active PFO, although regulation was not identical to that of LLO. The chromosomal PFO-expressing strain was characterized for intracellular growth and cell-to-cell spread. It escaped from a host cell vacuole with 64% efficiency compared with the wild type as determined by immunofluorescent staining of bacteria for F-actin, a marker for entry into the cytoplasm. In addition, it replicated intracellularly with a doubling time similar to that of the wild type for 5 h, after which growth was aborted because of a cytotoxic effect on the host cell and influx of extracellular gentamicin. The chromosomal PFO strain was able to plaque in mouse L2 fibroblasts, but it did so at 20% efficiency compared with the wild type and the plaques were significantly smaller. Both strains expressing PFO were completely avirulent in mice. These results indicate that PFO can mediate escape from a host cell vacuole but cannot complement an hly deletion strain for virulence.
